

2025 Shoe Sales Performance Analysis

A data-driven look at how shoe sales performed, which markets generated the highest revenue, and where future growth opportunities may exist.

Project

2025 Shoe Sales
performance Anlysis

Audience

Business Stakeholder

Prepared by

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Tools

Microsoft Excel

What we'll discuss.

01

Project Overview.

Project Overview presents the project's background, objectives, dataset, and key questions to provide context for the analysis

02

Data Cleaning.

Handling missing values, handling outlier data, removing duplicates, correcting data types, and ensuring data quality before analysis

03

Sales Performance Analysis.

Analyzing revenue trends, transaction patterns, and key business drivers

04

Dashboard Development.

Creating interactive visualizations to analyze performance, identify trends, and support business decision-making

05

Key Insight & Recommendations

Presents key findings and provides actionable recommendations based on data analysis to support business decision-making and business growth

Project Overview.

Project Background

In 2025, Company X conducted over 1,000 shoe sales transactions. As competition in the footwear industry continues to grow, understanding the behavioral patterns and performance dynamics of sales has become increasingly important. This project analyzes sales data to evaluate weekly/monthly sales dynamics, assess the impact of brand and product segments on performance, and generate actionable insights to help management navigate sales volatility and support data-driven decision-making

Data Source

<https://www.kaggle.com/datasets/syedaeman2212/shoes-sales-dataset>

Key Questions

- 01
- How is the monthly sales performance trend?
- 02
- Which factors have the greatest impact on revenue performance?
- 03
- Which country, channel, brands and products generate the highest revenue?
- 04
- What business strategies can improve overall sales performance?

Dataset Overview

Sale_ID	: Unique transaction ID
Date	: Transaction date
Brand	: Shoe brand
Shoe_Type	: Product category
Color	: Product color
Country	: Customer location
Sales_Channel	: Sales channel
Price_USD	: Product price
Units_Sold	: Quantity sold
Revenue_USD	: Total revenue

Data Cleaning.

	<u>Data Issue</u>	<u>Action</u>	<u>Impact on Analysis</u>
01	Identify Missing Value for each variable	Removed duplicate transactions based on unique Transaction IDs	Prevents revenue overstatement and skewed counts
02	Format Inconsistencies	Standardized date formats, naming conventions (Brand/Channel), and data types.	Enables seamless aggregation and dashboard integration
03	Missing Values	Handled missing fields via category-based imputation / record removal.	Ensures 100% data completeness without introducing bias.
04	Outliers & Anomalies	Filtered extreme transaction values using standard IQR / Z-Score thresholds	Normalizes data distribution for accurate statistical testing

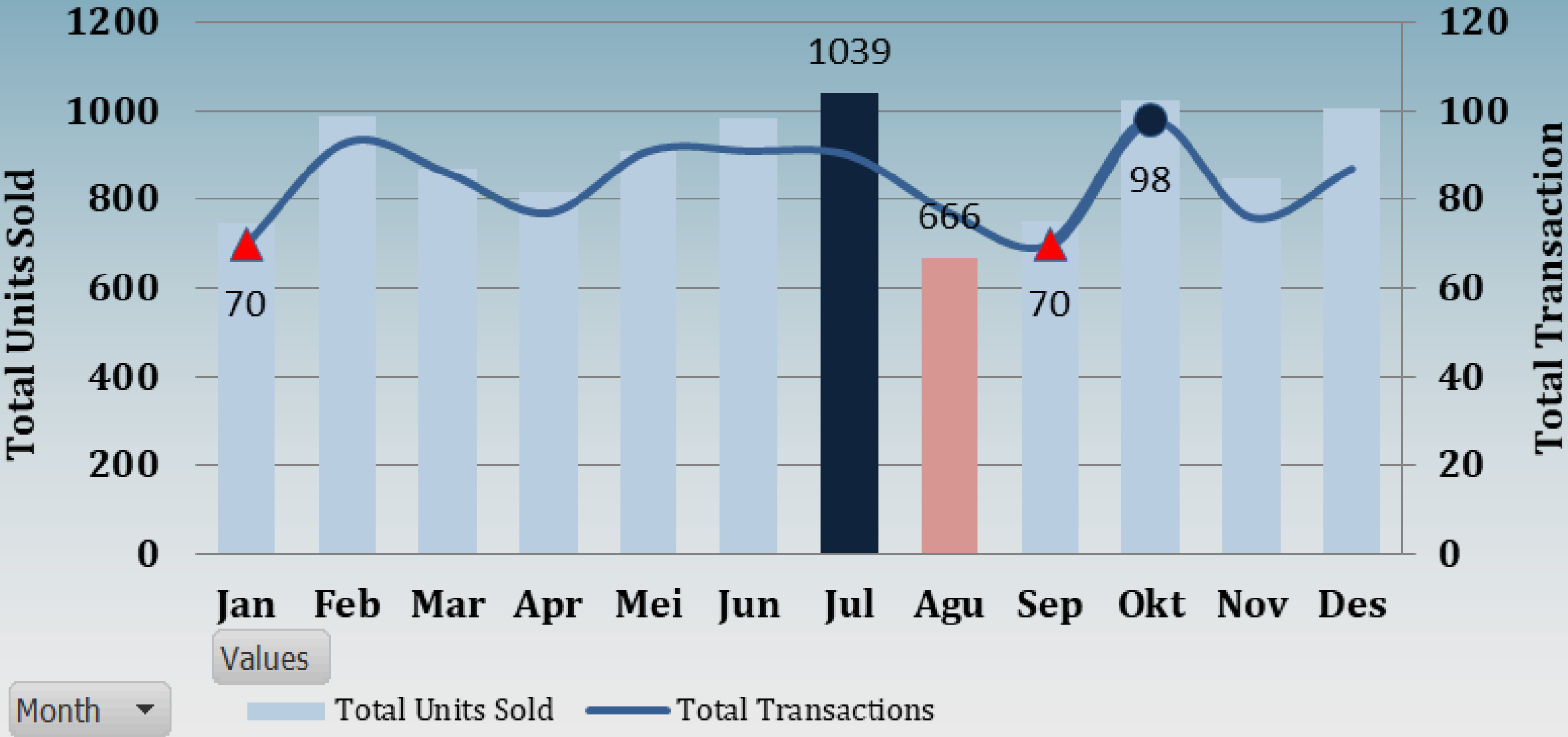
How do transactions trend over time?

10.651

Shoe Sold

1006

Transactions



October recorded the highest total transaction volume and a high number of units sold, **indicating stronger customer purchasing activity during the observed period.**

August recorded the lowest total units sold, accompanied by a decline in total transactions compared to the previous month. **This suggests weaker sales activity during the period.**

The company’s sales performance is dynamic, with two major peak periods in July and October, followed by the lowest point in August. **Transaction volume shows a linear relationship and serves as the primary driver of the total number of shoes sold.**

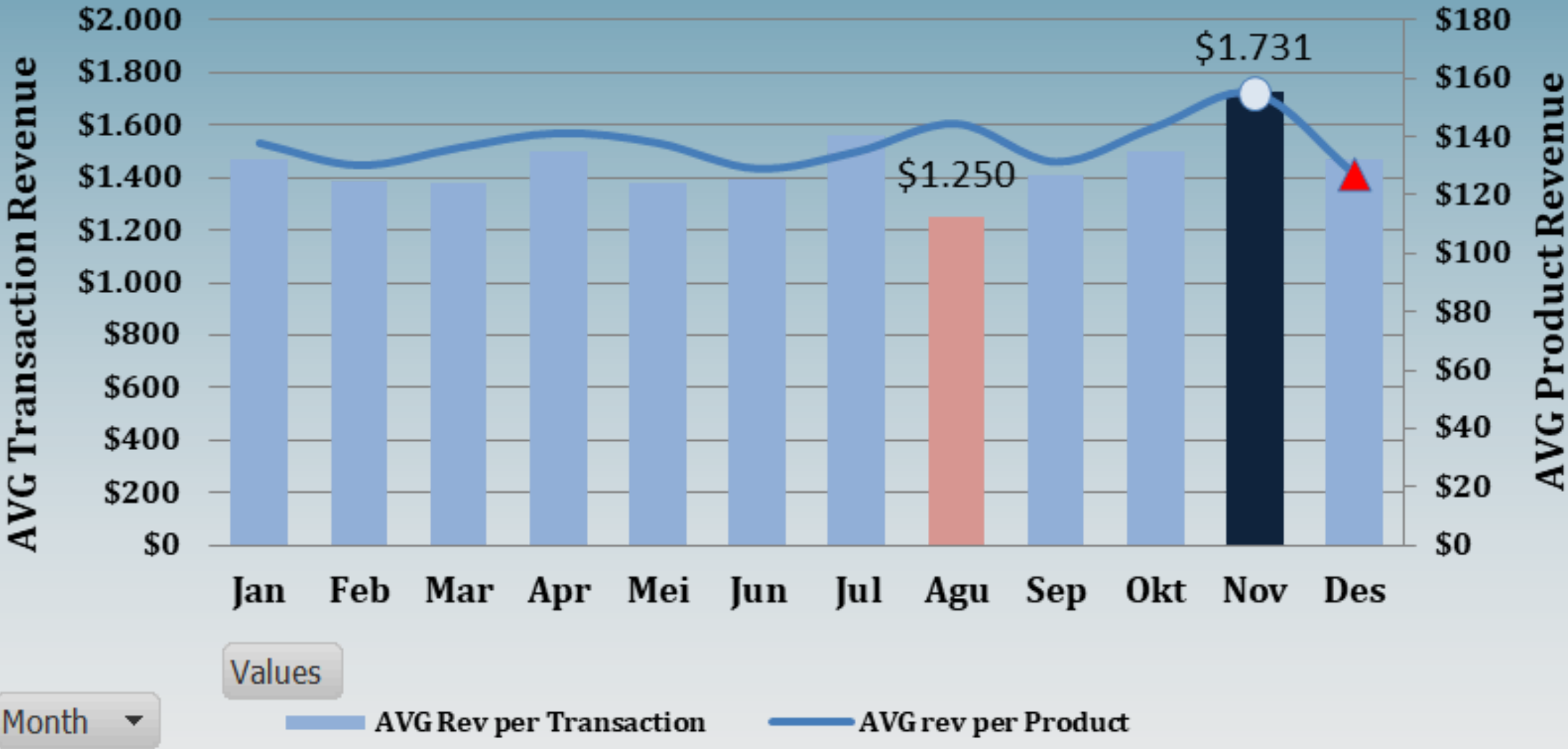
How do the monthly AVG revenue per transaction and product trend over time?

\$137

AVG Rev per Product

\$1.451

AVG Rev per Transaction



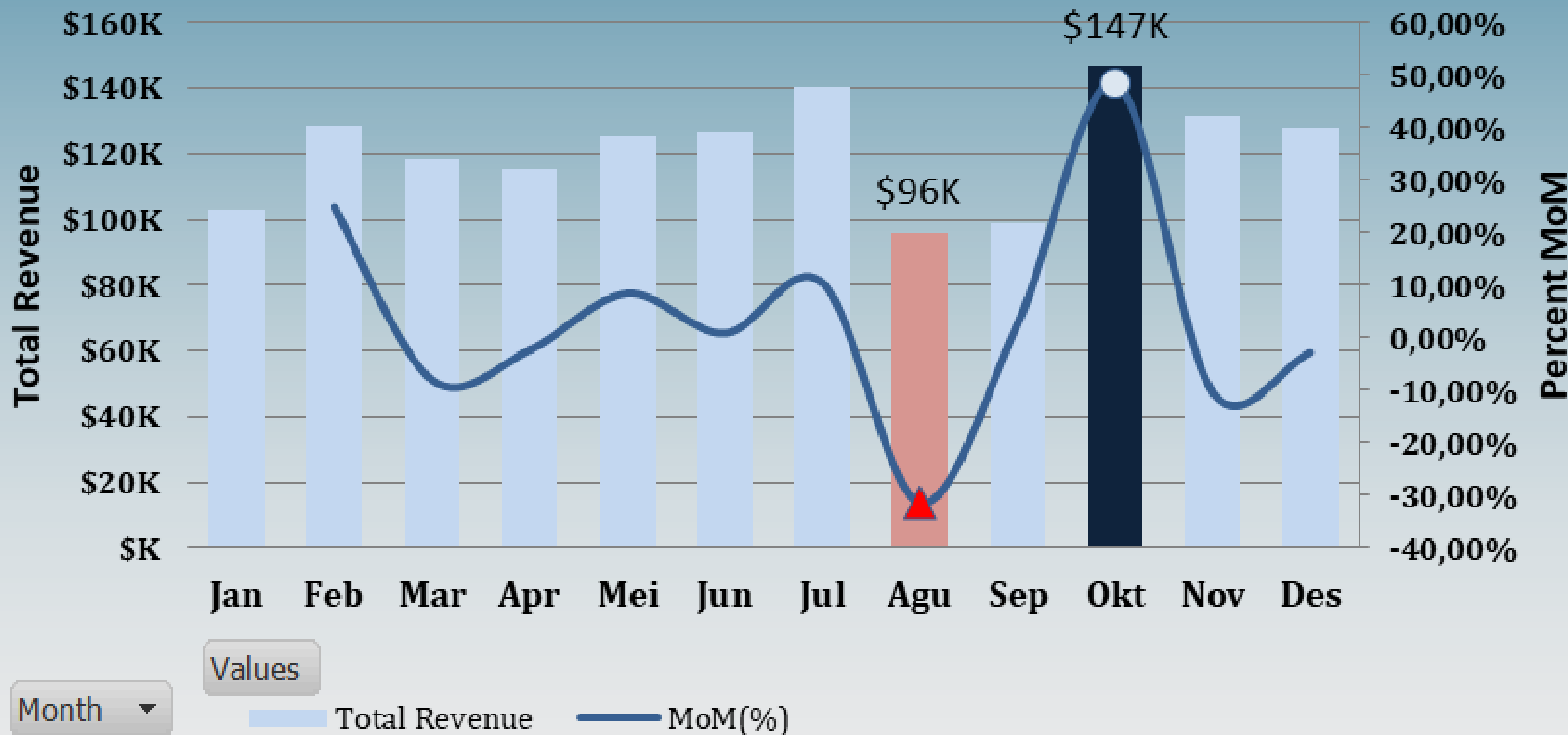
November recorded the highest across all months. However, both total transactions and units sold decreased compared to the previous month. This suggests that customers in November tended to purchase higher-value products while buying fewer items, resulting in a higher average transaction value despite lower sales volume

In August, average revenue per product was high while average transaction value was the lowest. Combined with the sharp decline in both transaction volume and units sold, this suggests that customers made fewer purchases overall but tended to purchase higher-value products

Total Revenue

\$1.459.848,63

How does revenue perform each month?



- The revenue trend and month-over-month (MoM) growth show a volatile but overall upward pattern, indicating growth with inconsistent performance. Although successive peaks increased over time, fluctuating troughs suggest that baseline performance remained unstable.
- The data also shows recovery patterns after periods of decline, with stronger rebounds but inconsistent low points.

Peak Month



October achieved the highest revenue performance This performance appears to have been driven by a combination of high transaction activity, increased sales volume, and a relatively high average revenue per transaction.

Lowest Month



August recorded the lowest total revenue and MoM growth, driven by declines across most key performance metrics, except for average revenue per product. This suggests that lower revenue performance was mainly associated with reduced sales volume and transaction activity.

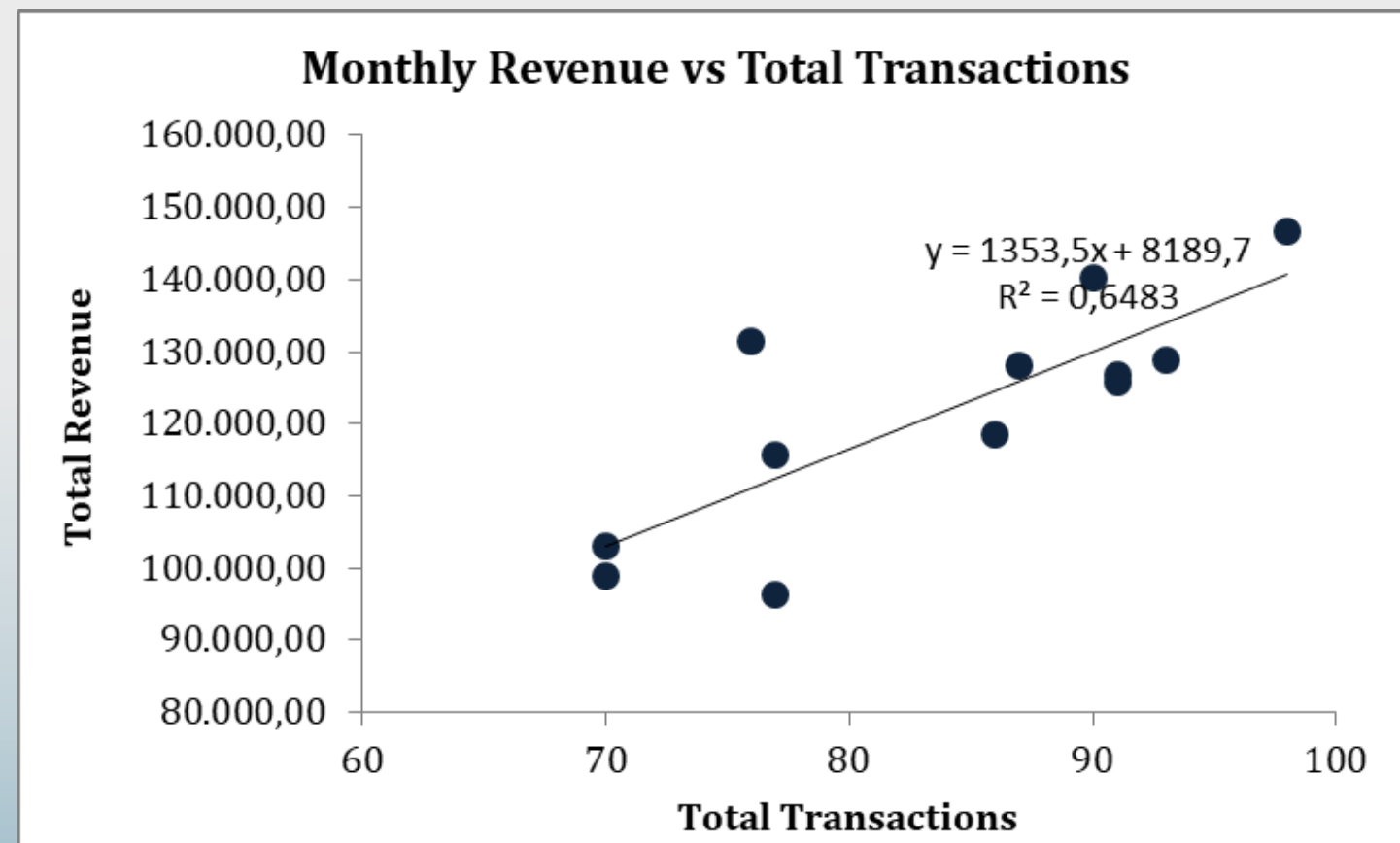
How do monthly sales performance trends affect business performance?

Business Concern: “Revenue Performance Variability

Although monthly revenue shows an overall upward trend, revenue and month-over-month (MoM) growth fluctuate considerably throughout the year. These **fluctuations reduce revenue predictability and make it more challenging to plan inventory, marketing activities, and business resources effectively**. Therefore, understanding the key drivers of monthly revenue performance is essential to support more stable and sustainable revenue growth

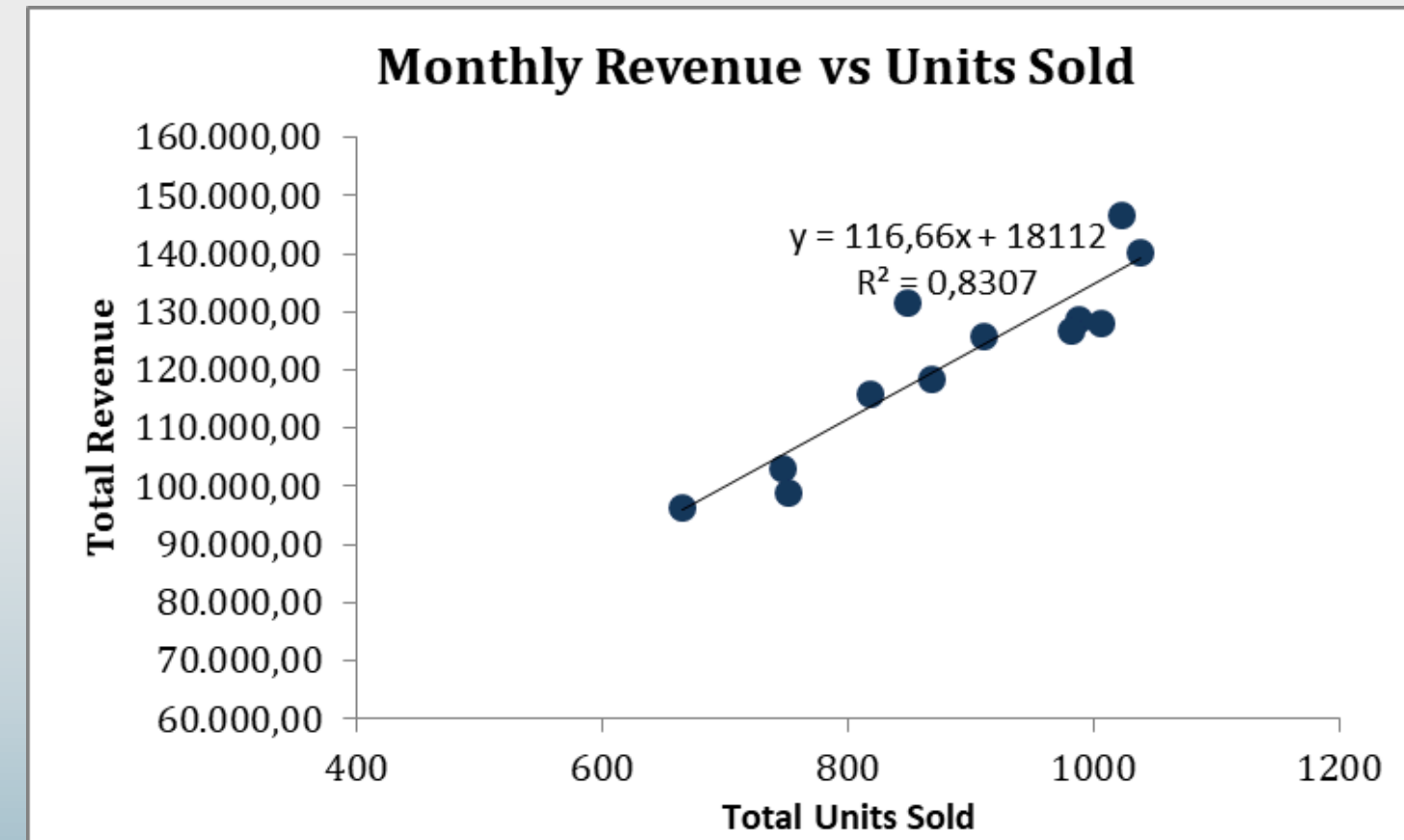
Business Question

“Which factors have the greatest impact on revenue performance?”



Metric	Result
R Square	0,648307664
F-statistic	18,43394346
Coefficients	1353,451386
P-value	0,00157755

- Total Transactions is a significant driver of monthly revenue performance, explains 64,8% of the variation in monthly revenue
- Each additional transaction is associated with an estimated increase of 1,353.45 in revenue.



Metric	Result
R Square	0,830660377
F-statistic	49,0529246
Coefficients	116,656425
P-value	0,00003699

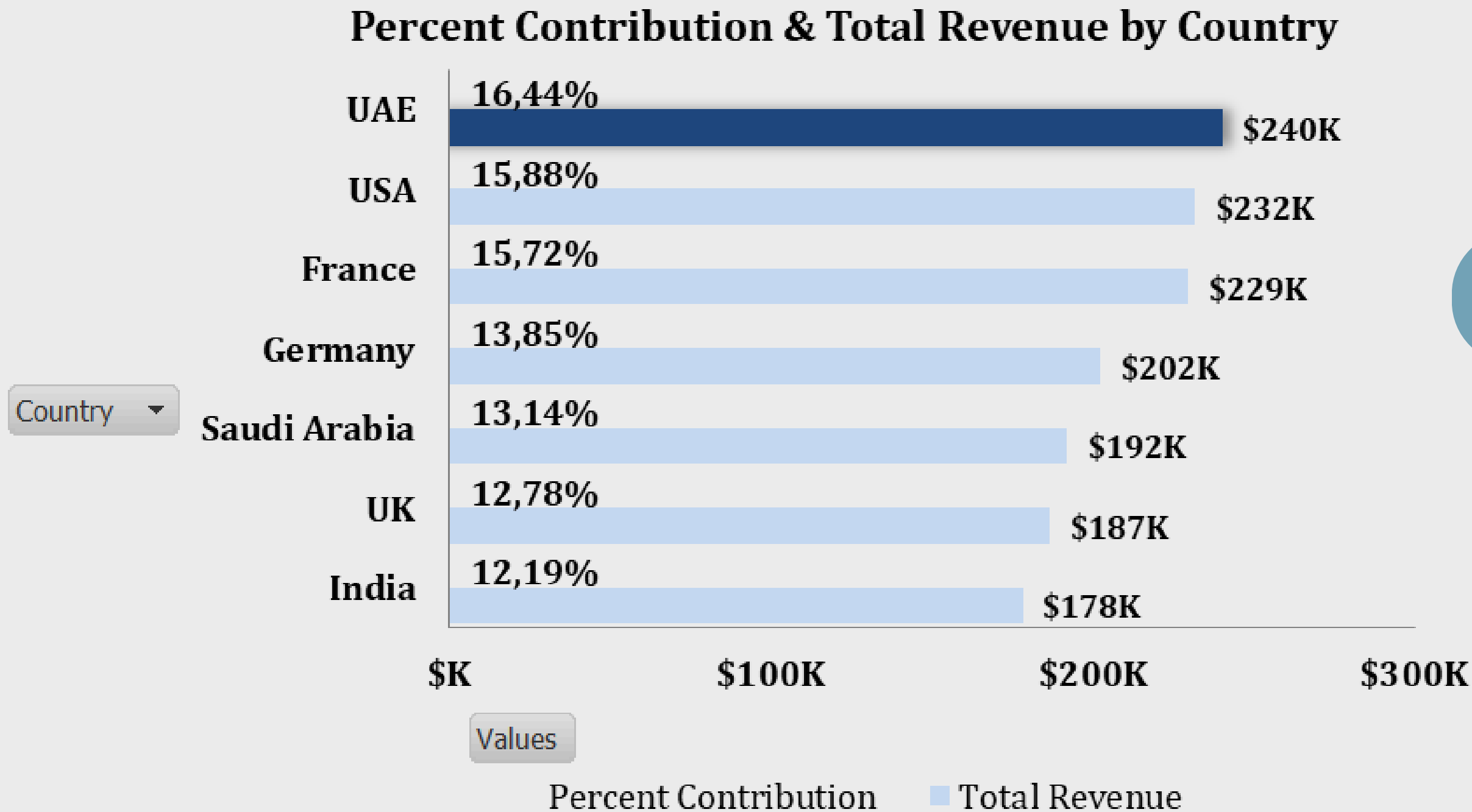
- **Units Sold is the strongest factor of monthly revenue performance.** The regression model explains 83.1% of the variation in monthly revenue
- Each additional unit sold is associated with an estimated 116.66 increase in revenue

What factors influence the number of units sold?

Summary statistics (Qualitative data):					Regression of variable Units_Sold:				
Variable	Categories	Counts	Frequencies	%	Variable	Categories	Counts	Frequencies	%
Brand	Adidas	165	165	16,402	Country	France	154	154	15,308
	New Balance	163	163	16,203		Germany	137	137	13,618
	Nike	144	144	14,314		India	146	146	14,513
	Puma	172	172	17,097		Saudi Arabia	128	128	12,724
	Reebok	165	165	16,402		UAE	155	155	15,408
	Skechers	197	197	19,583		UK	134	134	13,320
Shoe_Type	Boots	177	177	17,594	Sales_Chann	USA	152	152	15,109
	Casual	164	164	16,302		Mall	333	333	33,101
	Formal	177	177	17,594		Online	334	334	33,201
	Running	159	159	15,805		Retail Store	339	339	33,698
	Sneakers	179	179	17,793					
	Sports	150	150	14,911					
Color	Black	167	167	16,600					
	Blue	175	175	17,396					
	Green	176	176	17,495					
	Grey	178	178	17,694					
	Red	167	167	16,600					
	White	143	143	14,215					
					Goodness of fit statistics (Units_Sold):				
					Observations				
					Sum of weights				
					DF				
					R ²				
					Adjusted R ²				
					MSE				
					RMSE				
					MAPE				
					DW				
					Cp				
					AIC				
					AICC				
					SBC				
					PC				

- This data indicates that factors such as brand, country, color, shoe type, and sales channel **have a very minimal impact on units sold.**
- Therefore, the increase in units sold is likely driven by external factors not captured in the current product attributes, such as promotional discounts, special marketing campaigns, seasonal trends, or bundled offers

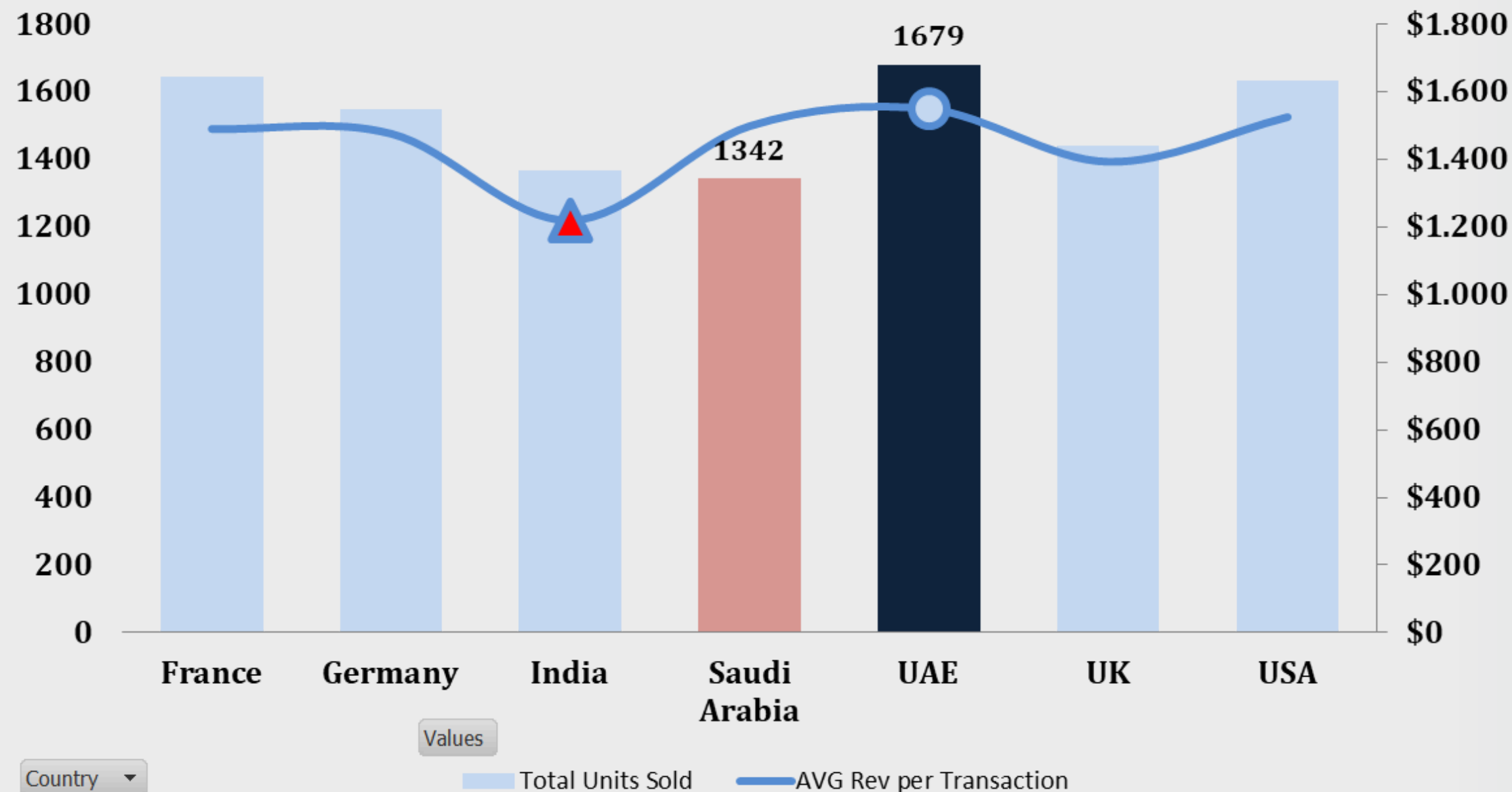
Which country generated the highest total revenue?



“The UAE generated the highest revenue among all countries”

The revenue contribution was only slightly higher than that of the other countries, ranging from 12.19% to 16.44%. This indicates a well-balanced market portfolio, where overall **business performance is supported by multiple markets rather than relying heavily on a single country**

How does sales performance trend across each country?



- **UAE recorded the highest values across key sales metrics**, including revenue, transaction volume, units sold, and revenue efficiency metrics. However, the differences compared with the other markets were relatively small, indicating a broadly similar level of performance across countries
- **Saudi Arabia** achieved strong revenue efficiency despite lower sales volume, **suggesting higher customer spending per transaction**
- In contrast, **India** recorded the lowest average revenue per transaction and the lowest total units sold. **indicating weaker customer demand and lower purchasing value compared to other markets**

Despite the similar overall performance, the underlying revenue drivers varied across markets. Different combinations of transaction volume, units sold, and average transaction value contributed to revenue generation, suggesting distinct customer purchasing patterns in each market

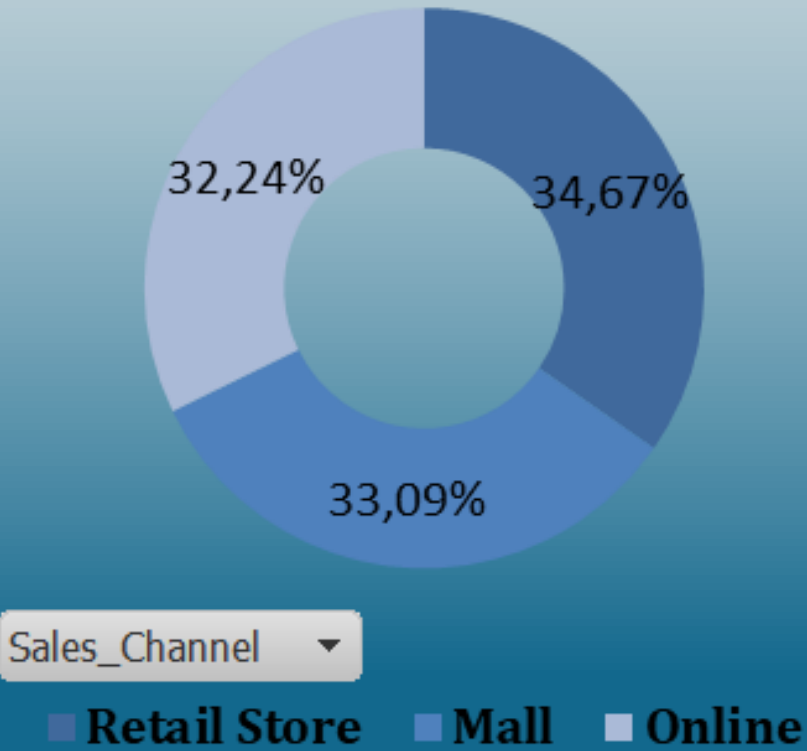
How does sales performance trend across each channel?

Channel	Total Revenue	Total Units Sold	Total Transaction
Retail Store	\$506,1K	3597	339
Mall	\$483,1K	3533	333
Online	\$470,7K	3521	334

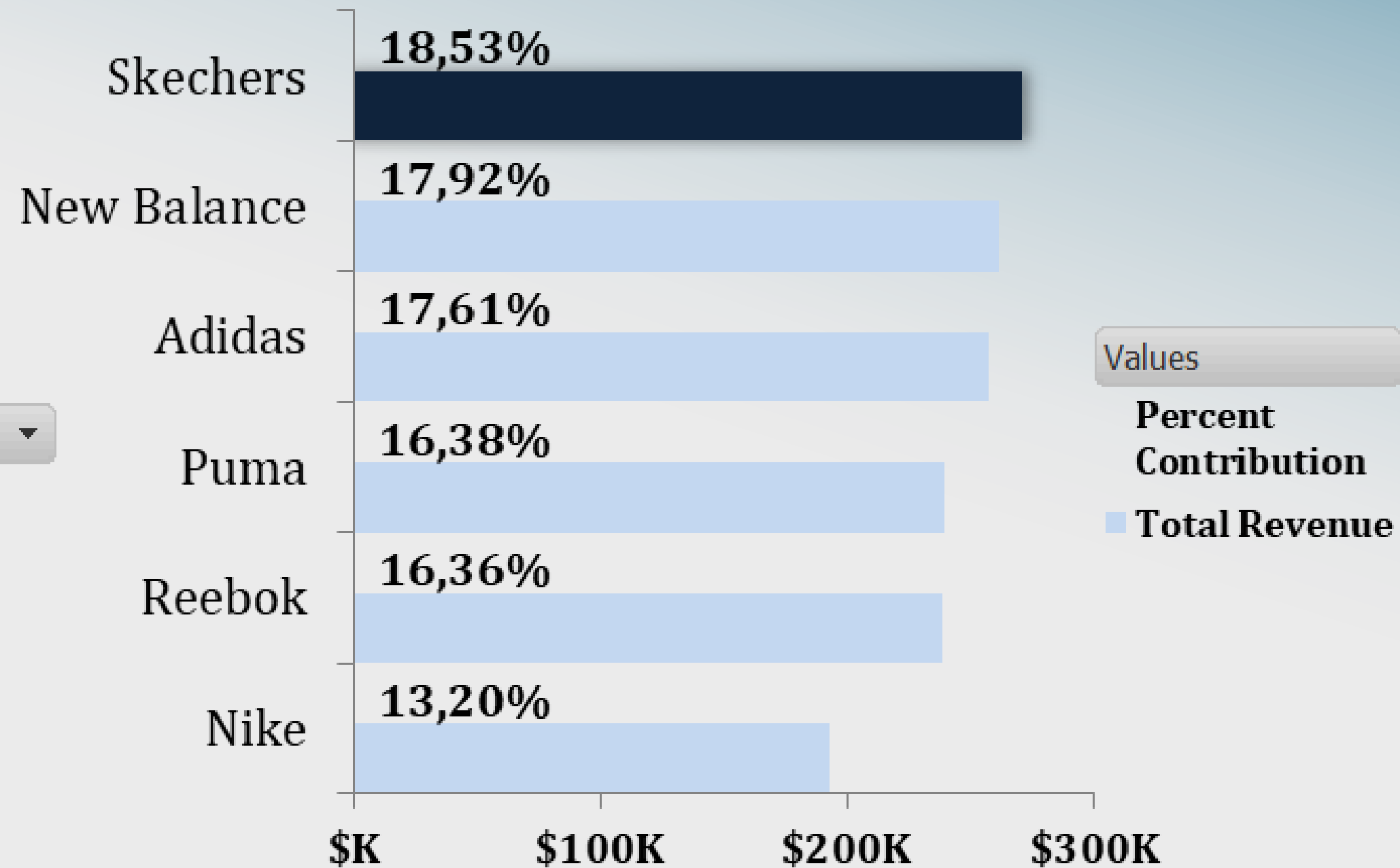
“Retail store generated the highest revenue among all sales channels”

Retail stores generated the highest revenue among all sales channels, although the performance gap across channels was relatively small. indicating that **all channels contributed similarly to overall revenue performance.**

Percent Revenue Contribution



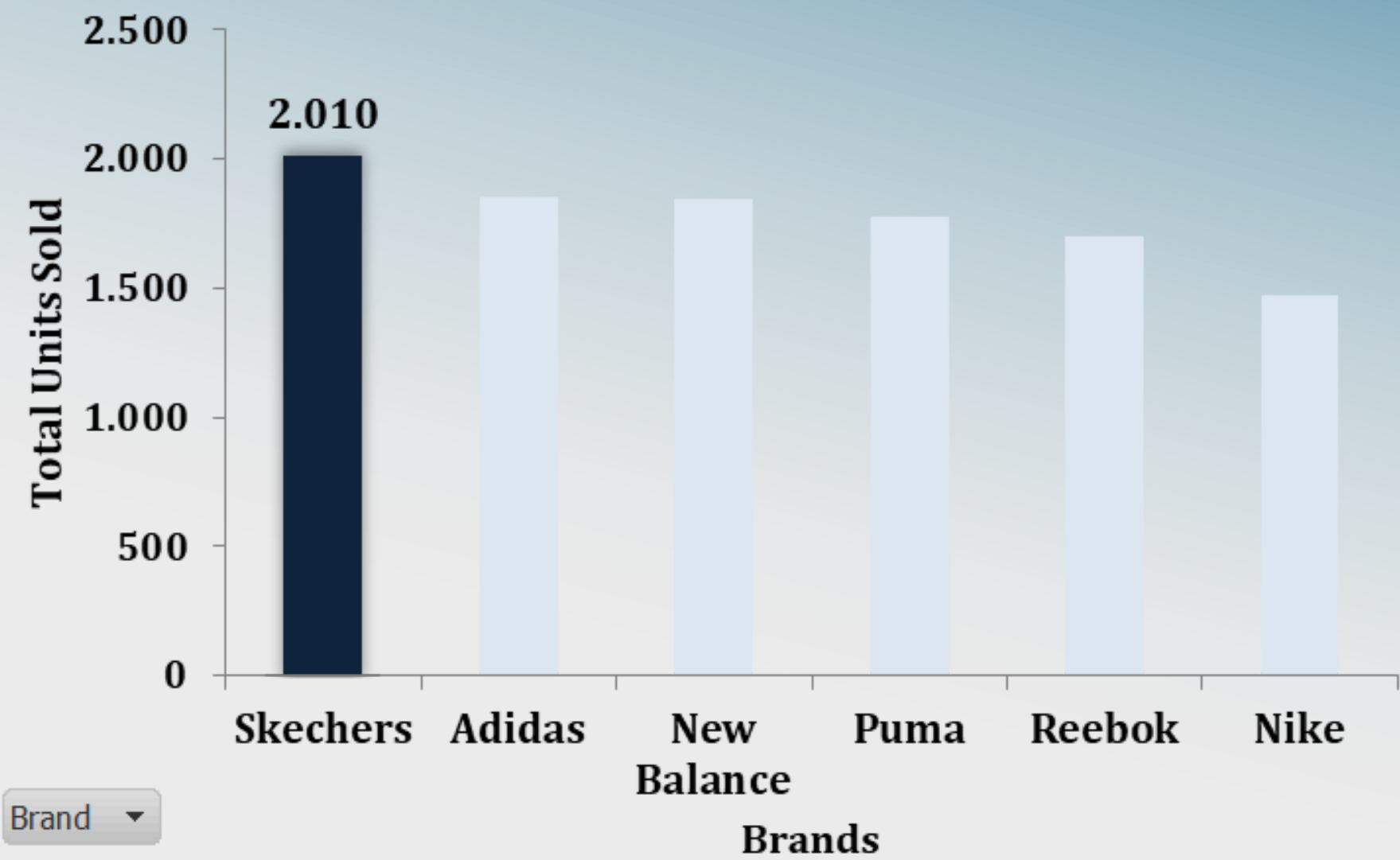
Which Brand Generated the Highest Revenue?



“Skechers generated the highest revenue among all brands”

The revenue contribution across brands is relatively balanced, with the top-performing brand contributing 18% of total revenue. However, the small gap between the highest and lowest contributors suggests that revenue is distributed relatively evenly across brands. **This indicates a well-balanced brand portfolio, where overall business performance is supported by multiple brands rather than relying heavily on a single brand.**

How does sales performance trend across each brand?



Brands	AVG per Product
New Balance	\$142
Reebok	\$140
Adidas	\$139
Puma	\$135
Skechers	\$135
Nike	\$131

“Most brands generated revenue primarily through sales volume, with higher units sold becoming the main driver of revenue performance.”

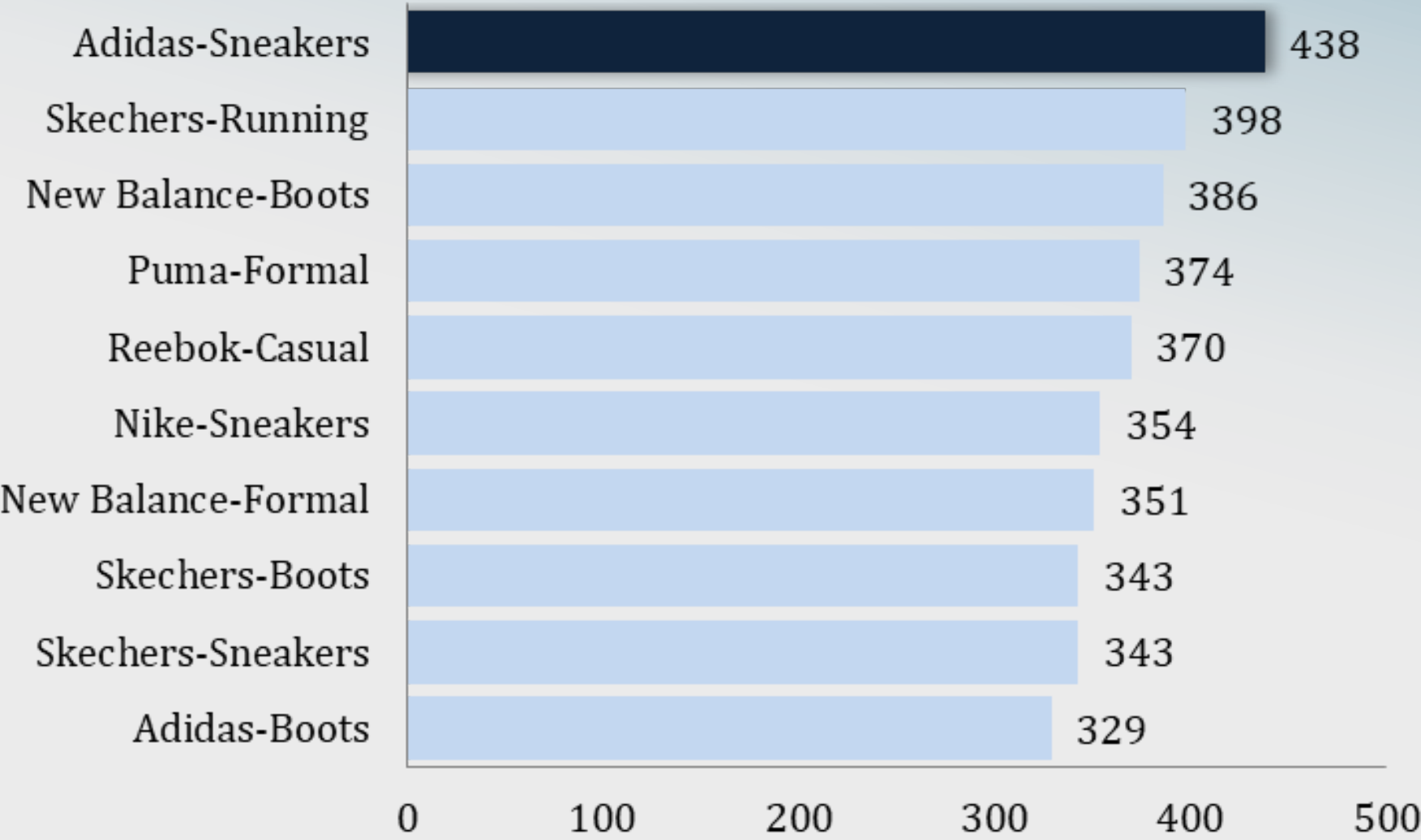
Skechers recorded the highest revenue among all brands, supported by its strong sales volume despite having a relatively lower average product value

Nike suggesting weaker overall sales performance compared to other brands.

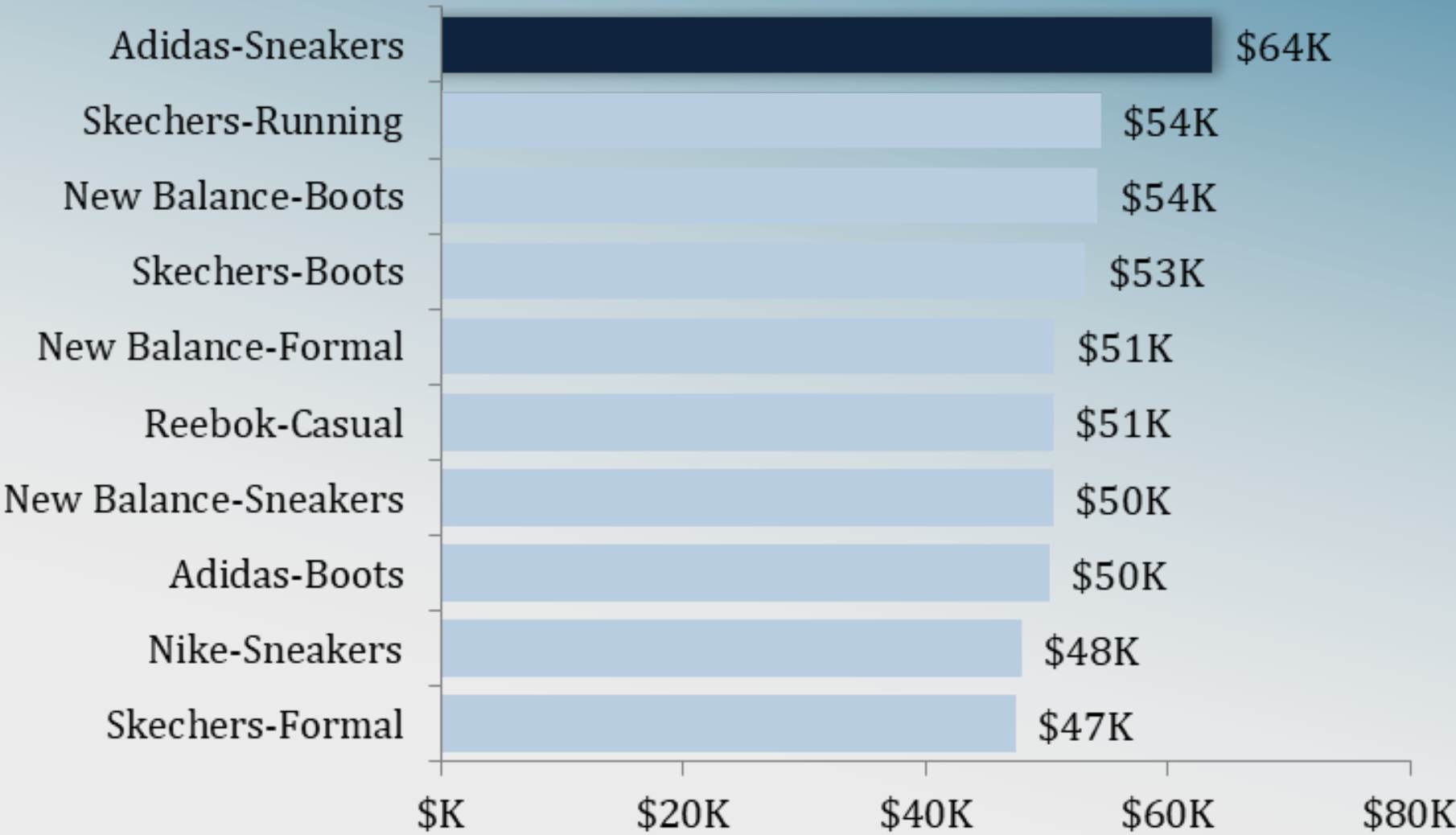
New Balance showed a different pattern, achieving strong revenue contribution through a combination of high unit sales and higher average product value

Which Products Were Purchased the Most and Generated the Highest Revenue?

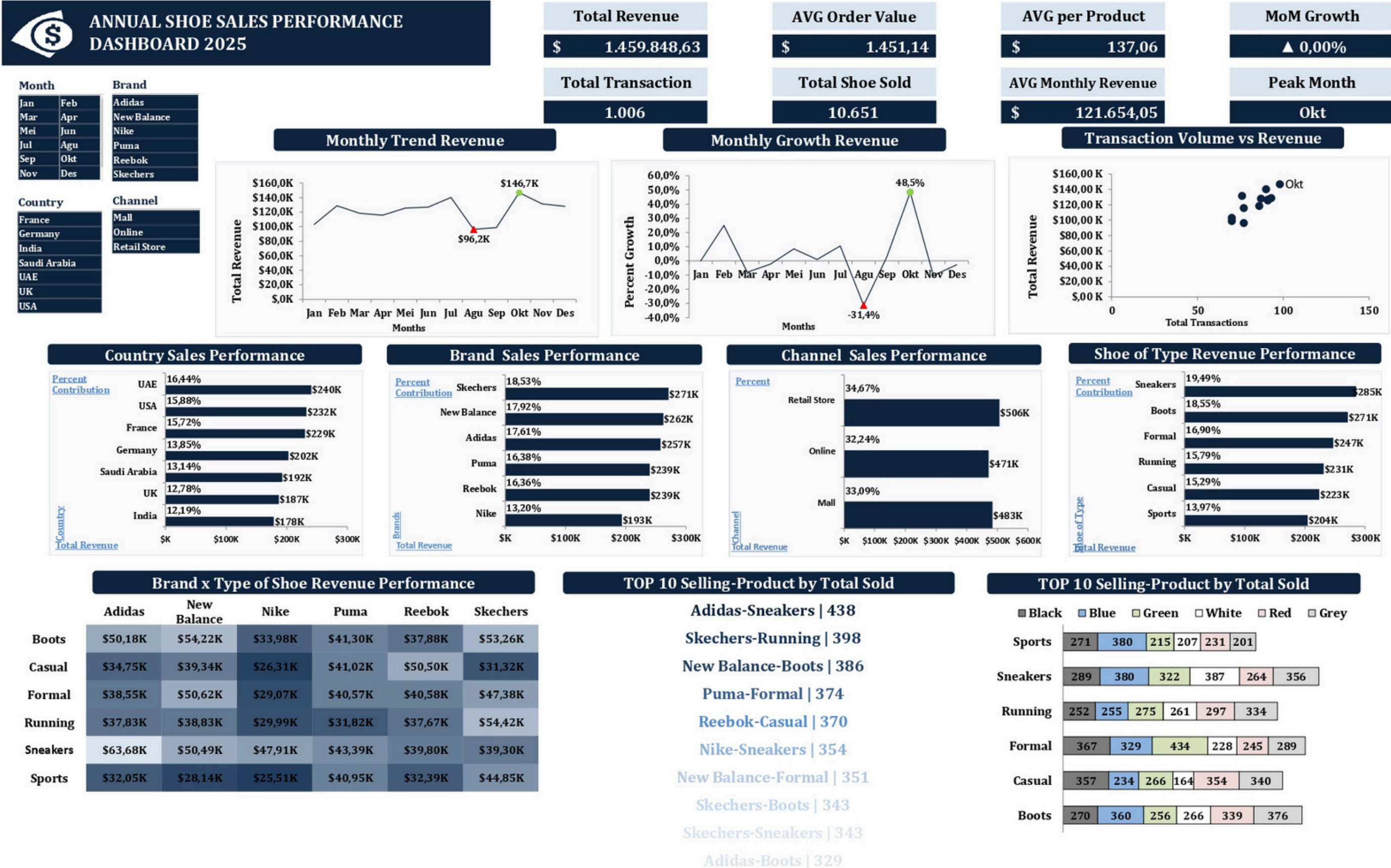
Top 10 Best-Selling Products by Units Sold



TOP 10 Poducts by Total Revenue



DASHBOARD REPORT



Key Insights Driving Revenue Performance and Growth Opportunities

Revenue Performance Shows Monthly Fluctuation

Revenue performance fluctuated throughout the year, with October achieving the highest revenue while August recorded the lowest performance

Channel and Brand Contributions are Relatively Balanced

Revenue contribution was distributed relatively evenly across sales channels and brands, with no single category dominating performance

Regional Performance Differs Mainly Due to Sales Volume and Transaction Value

Market performance varies across countries, with UAE generating the highest revenue contribution while India recorded the lowest performance

Total Unit Sold is the Main Revenue Driver

Revenue is strongly associated with sales volume, where higher units sold generally resulted in higher revenue performance.

Turning Insights into Actionable Business Strategies

Focus Area : Q3 Low-Season Recovery (August Revenue Decline)

Strategic Objective : Recover sales volume decline during low-performing periods by stimulating customer demand.

Strategic Action : Launch a targeted Back-to-School / Late Summer campaign before the historical August revenue decline to stimulate customer demand.

Product Focus :

Prioritize promotional efforts on historically high-performing SKUs, particularly Adidas-Sneakers, while introducing bundle promotions that pair them with lower-selling shoe products to help improve the sales performance of underperforming SKUs.

Offer Mechanism :

Test a bundle promotion (ex : Buy 2, Get 20% Off Second Pair) to encourage multi-item purchases and increase units per transaction

Turning Insights into Actionable Business Strategies

Focus Area : Regional Market Optimization

Strategic Objective : Optimize regional market performance through market specific growth strategies

Strategic Action :

India :

- Launch localized marketing campaigns featuring historically high-performing products to increase customer demand and purchase frequency.
- Test cross-selling bundle offers (ex : Footwear + Accessories or Shoe Care Kits) to encourage higher transaction values.
- Review product assortment and pricing strategy to better align with local customer purchasing behavior

UAE & Saudi Arabia :

- Prioritize premium and higher-priced product offerings in UAE and Saudi Arabia to align with stronger customer purchasing power.
- Ensure adequate availability of top-performing and premium SKUs during high-demand periods to prevent missed sales opportunities.
- Introduce exclusive product collections or limited-time launches to encourage repeat purchases from high-value customers.

Turning Insights into Actionable Business Strategies

Focus Area : Omnichannel Efficiency Strategy

Strategic Objective : Capitalize on balanced channel distribution while maximizing profitability

Strategic Action :

Maintain Balanced Channel Stocking :

Maintain a balanced product assortment across Online, Mall, and Retail channels to support consistent revenue performance

Monthly Multi Metric Performance Tracking :

Monitor channel performance regularly using Revenue, Units Sold, and Transactions to identify changes in customer purchasing patterns

Channel Tailored Promotional Testing:

Optimize channel specific promotional activities based on historical sales performance.

Turning Insights into Actionable Business Strategies

Focus Area : Q4 Peak Season Capitalization

Strategic Objective : Maximize high spending consumer behavior in October–November.

Strategic Action :

Prepare top-performing product availability before peak periods by monitoring historically high-performing product combinations and ensuring sufficient product availability to capture customer demand.

Implement cross-selling and upselling strategies during Q4 (ex : footwear + accessories bundles) to increase units per transaction and Average Revenue per Transaction.